Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/07/2006
PI Name:	Pfefferle, Lisa Ph.D.		
Project Title:	Small-scale, Multipurpose Catalytic Reactor for Combustion, Fuel Processing and Product Synthesis in Space		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	06520	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	01/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Dietrich, Daniel L	**Contact Phone:**	216-433-8759
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (10/06)		
Key Personnel Changes/Previous PI:	Professor Haller at Yale has provided significant input for the catalyst optimization.		
COI Name (Institution):	Haller, Gary Lee (Yale University)		
Grant/Contract No.:	NNC04GA19G		
Performance Goal No.:			
Performance Goal Text:			
	The tasks completed this year include the development of a catalyst and process for the direct conversion of methane to formaldehyde and optimization of a catalyst for the conversion of carbon dioxide to methane. The undergraduate student was involved with catalysts synthesis and optimization. For conversion of CO2 to methane we have developed a catalyst that shows over 96% selectivity for methane and recently showed how reactor design improves yield. This is significantly better than traditional catalyts. For the direct conversion of methane to formaldehyde we have developed a catalyst for high selectivity to formaldehyde allowing process optimization to yield a commercially viable process. This is an important process as it allows us to use methane as a precursor for producer chemical production. We have demonstrated over 56% selectivity at reasonable		

Task Description:	conversion, a significant achievement which would make this process commercially viable over the current two stage process (methane to methanol, then methanol to formaldehyde). Formaldehyde is a very large volume producer chemical so the new process when optimized is potentially valuable. Current commercial processes are multistep, not suited for use in a Mars environment and very waste intensive. A patent disclosure has been submitted. Our overall objective is to develop processes for producer chemical production in a Mars environment. We have been successful in engineering catalysts that are high selectivity for the desired products in the last part of the project we are further optimizing catalysts and focusing on reactor design. New catalysts and compact reactor designs have been demonstrated. Current processes are multistep and waste intensive. In particular the direct methane to formadehyde process is far less cumbersome and waste producing than traditional process and thus consistent with application in a Mars environment. Our process is made possible by our optimize vanadium incorporated MCM-41 catalyst. As noted below formaldehyde is a major producer chemical so this also has Earth applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our one step methane to formaldehyde process has significant environmental implications. Current processes are multistep and waste intensive. Our process (made possible by our optimize vanadium incorporated MCM-41 catalyst) is clean and require less capital investment. Formaldehyde is a major producer chemical.
Task Progress:	We have met the goals listed in our initial proposal. We have optimized at catalyst for the conversion of carbon monoxide to methane and used a short contact time reactor for the synthesis. We have also optimized a new catalyst for the conversion of methane to formaldehyde. Carbon Dioxide to Methane We have optimized a catalyst for conversion of carbon dioxide to methane in a Mars environment. The catalyst for this process is based on the incorporation of Nickel in an MCM-41 mesoporous silica framework in order to maintain high dispersion. Pretreatment in hydrogen resulted in the best yield. For 3% Ni incorporated Ni-MCM-41, 96% selectivity to the desired product (methane) was achived at a space time yield of 633gm/kgh. An interesting feature of our catalyst system is that selectivity increases as conversion increases. Because the selectivity increases with conversion we can go to higher temperatures in Ni-MCM-41 catalysts with smaller pores and this will give us even better yields. (The reducibility of the Ni in the MCM-41 decreases as pore size is decreased so small metal particles will be stable at higher temperatures). Methane to Formaldehyde The current state of the art for converting methane to formaldehyde involves cumbersome 2 and 3 step processes that would be difficult if not impossible to implement in space. We have developed a process that produces very high selectivity formaldehyde at yields making it a practical and economic process. A selectivity of 60% was achieved at a space hourly time yield of 1315g/kghr significantly higher than with any other catalyst system. The catalyst that allows these results is based on the incorporation of Vanadium in an MCM-41 mesoporous silica framework in order to maintain high dispersion and optimize the stabilization of the isolated vanadium species that is most selective for formaldehyde production. The very high yield to fomaldeyde allowed a simplified one step process for the production of formaldehyde from partial oxidation of methane to be developed and optimized. A paper was submitted and in the final review stages on the novel application of statistical multivariate analysis and prediction through the multi-regression model for the optimization of the processing parameters in the complex reaction process. This approach is advantageous, especially when experimental evaluation and optimization of a process is time consuming and expensive. By carrying out a finite number of experiments, statistical modeling in this work shows reasonable good prediction ability in terms of methane conversion, formaldehyde selectivity, and space time yield. Moreover, not only are the synergistic effects between reaction parameters revealed, but also a comprehensive understanding of the whole production process over variation of the all reaction parameters (within the phase space represented by the experimental grid) are allowed. The final process is simpler and less waste producing than the convential multistep.
Bibliography Type:	Description: (Last Updated: 06/12/2006)
Articles in Peer-reviewed Journals	Du G, Lim S, Yang Y, Wang C, Pfefferle L, Haller G. "Catalytic performance of vanadium incorporated MCM-41 catalysts for the partial oxidation of methane to formaldehyde." Appl Catal A Gen. 2006 Mar 21;302(1):48-61. , Mar-2006
Articles in Peer-reviewed Journals	Du G, Lim S, Yang Y, Wang C, Pfefferle L, Haller G. "Statistical Design and Modeling of the Process of the Partial Oxidation of Methane and Prediction of Formaldehyde Production." Applied Catalysis A, in review, June 2006. , Jun-2006